W-F-441C
October 29, 1971
SUPERSEDING
Int. Fed. Spec. W-F-00441B
August 12, 1970, and
Fed. Spec. W-F-441A
September 23, 1964

FEDERAL SPECIFICATION

FLATIRONS, ELECTRIC, (DRY) (EXCLUDES TRAVEL IRON)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers alternating current electric flatirons with adjustable automatic temperature control.

1.2 Classification.

1.2.1 Class. Flatirons shall be of the following class as specified in the invitation for bids (see 6.2):

Class 1 (Weight - 2.5 pounds minimum to 4.5 pounds maximum).
Class 2 (Weight - Over 4.5 pounds to 5.5 pounds maximum).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification the extent specified herein.

Federal Specifications:

QQ-C-320 - Chromium Plating (Electrodeposited).
QQ-N-290 - Nickel Plating (Electrodeposited).
PPP-B-636 - Box, Fiberboard.
PPP-C-843 - Cushioning Material, Cellulosic.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

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(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Underwriters' Laboratories, Inc., (UL) Standards:

No. 141 - Electric Flatirons and Ironing Machines.
No. 498 - Attachment Plugs and Receptacles.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, IL 60611; 333 Pfingston Road, Northbrook, IL 60062; 1285 Walt Whitman Road, Melville, L.I. NY 11746; 1655 Scott Boulevard, Santa Clara, CA 95050.)

American National Standards Institute, Inc.

C70.1-1953 - Household Automatic Electric Flatirons.

(Application for copies should be addressed to the United States of American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

National Motor Freight Traffic Association, Incorporated, Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 202 Union Station, 516 W. Jackson Blvd., Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Representative sample. Unless otherwise specified (see 6.2), before production is commenced, a sample flatiron shall be submitted or made available to the contracting officer or his authorized representative for examination and tests in accordance with 4.2. Approval of the representative sample authorizes the commencement of production but does not relieve the supplier of the responsibility for compliance with the detailed provisions of this specification. The representative sample shall be manufactured utilizing the same production tooling to be used for the manufacture of the production items.

3.2 UL Certification. Each contractor shall submit to the contracting agency proof that the flatiron he proposes to supply under this specification conforms to the requirements of the Underwriters' Laboratories, Inc. Standards for Safety, 141, Electric Flatirons, and 498, Attachment Plugs and Receptacles. The label, or listing with re-examination, of the Underwriters' Laboratories, Inc. may be accepted as evidence that the flatiron conforms to this requirement. In lieu of the label, or listing with re-examination, the contractor may submit independent proof, satisfactory to the contracting agency, that the flatiron conforms to the applicable requirements of the published standards including the methods of tests of the Underwriters' Laboratories Standards for Safety 141, Electric Flatirons, and 498, Attachment Plugs and Receptacles.

3.3 Standards. Unless otherwise specified, the flatirons shall conform to the requirements of ANSI Standard C70.1-1953, UL Standards 141 and 498. Where ANSI and UL Standards conflict, the UL Standard shall take precedent.

3.4 Materials. The materials specified shall conform to the applicable specifications and requirements specified hereinafter. Materials not definitely specified shall be of the quality normally used for the purpose in commercial practice.

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3.5 Finish. Unless otherwise specified, (see 6.2), all outside metallic surfaces of the flatiron except the soleplate shall have a chromium over nickel plating, baked on enamel, or porcelain enamel finish to conform to the requirements of 3.5.1 through 3.5.3.

3.5.1 Ferrous metal. The nickel plating on ferrous metal shall conform to class 1, type IV as specified in QQ-N-290. The chromium plating shall conform to class 1, type I as specified in QQ-C-320.

3.5.2 Copper, brass, bronze. The nickel plating shall conform to class 1, type VII as specified in QQ-N-290. The chromium plating shall conform to class 1, type I as specified in QQ-C-320.

3.5.3 Enamel. Unless otherwise specified (see 6.2) the color of baked on enamel or porcelain enamel finishes shall be white. The finish shall be of the quality normally used in commercial practice. At the end of the tests specified in 4.6 the test samples shall be examined; there shall be no evidence of chipping, cracking, discoloration, or scaling of the enamel finish (see 4.4).

3.6 Rating.

3.6.1 Rated voltage. The rated, normal operating voltage for the flatiron shall be 120 volts.

3.6.2 Rated watts input. The rated watts input of the flatiron shall be not less than 950 watts minimum nor more than 1250 watts maximum.

3.6.3 The wattage input to the flatiron shall be determined in accordance with 4.6.4 for conformance to the wattage marked on the flatiron. Failure to pass this requirement shall cause the entire lot to be rejected.

3.7 Construction.

3.7.1 Handle. The handle shall be made of phenolic or a similar high temperature composition. The handle shall be molded in one piece with the supporting bracket or shall be securely fastened to the handle bracket.

3.7.2 Soleplate.

3.7.2.1 Material. The soleplate shall be made of either aluminum or other alloys, or stainless steel overlay on aluminum or other alloys.

3.7.2.2 Area. The soleplate area shall not be less than 22 square inches.

3.7.2.3 Configuration. The soleplate shall have a normal point. There shall be a bottom nook on each side of the point or the soleplate shall have flared leading edges to serve the purpose of the button nooks.

3.7.2.4 Bottom surface. The bottom surface of the sole plate may have a bevel edge around the perimeter at the option of the manufacturer. The dimensions of the bevel edge shall not be greater than 1/2 inch wide by 0.060 inches deep when tested as specified in 4.6.1.1. The variation from flatness of the soleplate when placed against a straight-edge, shall be no greater than 0.030 inches when tested as specified in 4.6.1.2.

3.7.3 Safety stand. Each flatiron shall have a permanently attached safety stand or rest.

3.7.4 Weight. Class 1 flatirons shall weigh not less than 2.5 pounds nor more than 4.5 pounds Class 2 flatirons shall weigh over 4.5 pounds and no more than 6.5 pounds.

3.7.5 Thermostat. Each flatiron shall have a manually adjustable thermostat. The thermostat knob or lever shall be of such shape and so located that temperature adjustment can be made without the hand of the operator coming in contact with a hot surface. The thermostat control knob shall have an "OFF" position.

3.7.6 When the thermostat control knob is in the "OFF" position there shall be no current flow through the ammeter when tested in accordance with 4.6.5.

3.7.7 Cords. Each flatiron shall have a cord, not less than 6 feet in length, either permanently attached to or detachable from the flatiron. One end of the cord shall be equipped with an attachment plug, conforming to the Underwriter's Laboratories, Inc., Standard 498, for connection of the cord to the power supply.

3.7.7.1 Detachable cord. When a detachable cord is furnished, the flatiron shall be equipped with on-the-iron contact posts of corrosion-resisting material. One end of the detachable cord shall be equipped with a flatiron plug having sleeves of corrosion-resisting material to fit the contact posts on the iron.

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3.8 Speed of heating. Each flatiron shall attain a mean sole plate temperature of not less than 400°F. within 4 minutes after being connected to a power supply and tested as specified 4.6.2.

3.9 Cord-guard. Each cord shall have a cord-guard not less than 2 inches in exterior length. The cord-guard shall be located at the flatiron plug when the cord is detachable and at the flatiron when the cord is attached.

3.10 Marking. The flatirons shall be marked in accordance with the marking requirements of Underwriters' Laboratories, Inc., Standard 141, for electric flatirons and ironing machines.

3.10.1 Marking of the thermostat control fabric plate. The thermostat control fabric plate shall be marked to indicate settings for not less than four types of fabric or shall have alpha or numeric settings which are related to the proper fabrics marked on a separate cover plate. The fabric markings shall include Synthetics, Wool Cotton and Linen.

3.11 Workmanship.

3.11.1 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools, and shall completely fill the holes. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.11.2 Welding and brazing. All surfaces of parts to be welded or brazed shall be clean. Welding and brazing shall be accomplished in a manner which will prevent the occurrence of burn holes, cracks, fractures or incomplete fusion. All scale or flux deposits shall be removed from finished welds.

3.11.3 Electrical connections. Electrical connections shall be mechanically secured and welded or brazed; except connections on screw type terminals, which shall be mechanically secured.

3.11.4 End item. The end item shall be assembled in a manner to insure that all parts are in proper alignment and that all exposed surfaces are free of sharp burrs, slivers, pits, indentations, cracks, ridges, blow holes, or checks that may damage fabrics or injure the user.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Representative sample inspection. When a representative sample is required, it shall be examined in accordance with the defects in table I and tested in accordance with in 4.6.1 through 4.6.6.2.

4.3 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated.

4.3.1 Component and material inspection. In accordance with 4.1, the supplier is responsible for insuring that components and materials used are manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or, if none, in accordance with this specification.

4.3.2 Lot. A lot shall consist of all flatirons of the same class offered for acceptance at one time. The sample unit shall be one completely fabricated flatiron.

4.4 Visual examination. Examination of the end item shall be in accordance with the classification of defects in table I. The inspection level shall be level II and the acceptable quality level (AQL) 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

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TABLE I. Classification of defects

Examine	Defect	Classification	
		Major	Minor
Construction and Workmanship General (Applicable to Components and Assemblies).	Materials not as specified (3.4).	X	
	Finish not as specified (3.5).		X
	Enamel finish shows chipping, cracking, discoloration, or scaling (3.5.3).		X
	Handle not as specified (3.7.1).	X	
	Soleplate not as specified (3.7.2).	X	
	Safety stand not as specified (3.7.3).	X	
	Weight not as specified (3.7.4).		X
	Thermostat not as specified (3.7.5).		X
	Cord not as specified (3.7.7 or 3.7.7.1).	X	
	Cord-guard not as specified (3.9).	X	
	Riveted connections not in accordance with 3.11.1.		X
	Welded or brazed areas not in accordance with 3.11.2.	X	
	Electrical connections not as specified (3.11.3).	X	
	End item not in accordance with 3.11.4.	X	
Marking	Missing, incomplete, not legible (3.10).		X
	Thermostat control fabric plate not marked as specified (3.10.1).	X	

4.5 Testing of the end item. End item testing shall be performed on the flatiron in accordance with 4.6.1 through 4.6.6.4. The inspection level shall be S-2 with an AQL of 4.0 expressed in term of defects per hundred units.

4.6 Tests.

4.6.1 Variation of soleplate flatness.

4.6.1.1 Bevel edge. Place the soleplate of the flatiron on a true plane surface and with feeler gages determine the dimensions of the bevel edge for compliance with 3.7.2.4.

4.6.1.2 Soleplate surface less bevel edge. With the soleplate at eye level position, move a straight-edge across the bottom surface of the soleplate in a direction parallel to the longitudinal axis of the flatiron and again in a direction perpendicular to the longitudional axis of the flatiron. With a feeler gage, determine any variations of bottom surface of the soleplate for compliance with 3.7.2.4.

4.6.2 Speed of heating. Attach a disc type thermocouple, see 4.6.6.1, to the mid-point of the soleplate. Connect the flatiron to a nominal 120-volt, 60 hertz, 1-phase source metered to read input volts, input ampers, and input watts. Starting from an ambient temperature of 60°C ± 0°F (20°C ± 5°C), set the thermostat control at the highest temperature setting. The minimum time that it takes the soleplate to reach a temperature of 400°F (204°C) shall be noted for compliance with 3.8.

4.6.3 Determination of the hottest point. With the flatiron connected to the voltage circuit, separate the flatiron from the thermocouple. Place the flatiron on a sheet of white paper or other material until scorching begins. After removing the flatiron, darkening of the material indicates the temperature distribution over the soleplate. The hottest point is determined as the center of the darkest area.

4.6.4 Watts input. With the flatiron resting on it's stand or other stand, and still connected to the voltage circuit and the thermostat control set at the highest setting, note the metered watts and volts input reading. The metered watts input shall be equal to

$$Wm \left(\frac{Ve}{Vm}\right)^2$$

Where Wm is the marked rated wattage of the flatiron, Ve is the line input voltage, and Vm is the marked rated voltage of the flatiron (see 3.6.3).

4.6.5 Thermostat "OFF" position. With the flatiron connected to the voltage circuit, return the thermostat control knob to the "OFF" position. Note the reading of the ammeter for compliance with 3.7.6.

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4.6.6 Average soleplate temperature tests.

4.6.6.1 Temperature measurement method. Temperature measurements shall be made with thermocouples attached to a metal disc by means of a heat resistant adhesive or by thermocouples which are peened into the soleplate or welded to the soleplate. The holes into which the thermocouples are peened shall be approximately 1/32 (0.031) inch in diameter. The disc to which thermocouples are attached shall be reasonably smooth and may be coated with silicone oil to insure good contact.

4.6.6.2 Thermocouple placement. Four thermocouples, of a type described in 4.6.6.1 attached to separate potentiometers, shall be used. The flatiron shall be supported in a manner that the support will not contact the thermocouples. When disc type thermocouples are used, a three prong support may be used to allow the thermocouple discs to contact the soleplate with equal pressure. A thermocouple is applied at each of the following points: a) at the hottest point as determined in 4.6.3, b) at the midpoint of the soleplate, c) at a point on the longitudinal center line approximately 0.781 in. or 20 mm from the tip of the soleplate, and, d) at a point on the longitudinal center line approximately 0.781 in. or 20 mm from the base of the soleplate.

4.6.6.3 Lowest thermostat setting. Attach the thermocouples and support the flatiron as described in 4.6.6.2. Connect the flatiron to an electrical circuit in accordance with 4.6.2. Adjust the voltage so that the watts input is the same as marked on the flatiron. Determine the average soleplate temperature for the lowest thermostat setting in accordance with the procedures in ANSI Std. C70.1 (see 3.3).

4.6.6.4 Highest thermostat setting. Repeat the procedure in 4.6.6.3 except that the thermostat shall be set at the highest setting (see 3.3).

4.7 Examination of preparation for delivery. An examination will be made to determine that packaging, packing, and marking as required by section 5 of this specification are complied with. Defects will be scored as specified in table II. The sample unit shall be one shipping container fully prepared for delivery selected at random. The lot shall be the number of containers offered for delivery at one time. The inspection level shall be S-2 and the acceptable quality level shall be 6.5, expressed in terms of defects per hundred units.

TABLE II. Examination of preparation for delivery

Examination	Defect
Markings	Missing; incorrect; illegible; of improper size, location, sequence or method of application.
Materials	Any nonconforming component; component missing, damaged or otherwise defective.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, inadequate stapling, bulging, or distortion of containers, strapping or sealing inadequate improper or missing; closure improperly affixed.
Weight	Net weight exceeds specification requirements.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A or B. Each flatiron shall be wrapped in cushioning material conforming to type I or II, class B of PPP-C-843. The flatiron shall then be placed in a fiberboard box conforming to type CF or SF, class Domestic, of PPP-B-636. The flatiron shall be immobilized within the fiberboard box with fiberboard inserts of the same material as the box. Each box shall be securely closed in accordance with method II of the appendix of the box specification.

5.1.2 Level C. Flatirons shall be packaged in accordance with the industry's practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

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5.2.1 Level A. Flatirons, packaged as specified in 5.1, shall be packed in a snug fitting fiberboard shipping container conforming to V3c or V3s, of PPP-B-636. The weight of contents of each shipping container shall not exceed 65 pounds. Each shipping container shall be closed, waterproofed by means of tape, and reinforced with flat steel strapping, or tape banding in accordance with method II of the appendix of the container specification.

5.2.2 Level B. Six (6) class 2 or twelve (12) class 1 flatirons packaged as specified in 5.1 shall be packed in a close fitting box conforming to PPP-B-636, class domestic. The box shall be closed in accordance with method II of the appendix to PPP-B-636.

5.2.3 Level C. The flatirons packaged as specified in 5.1 shall be packed in containers to assure carrier acceptance and safe arrival at destination in compliance with Uniform Freight Classification Rules or National Motor Freight Classification Rules.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, interior and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military requirements. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The flatirons covered by this specification are for general use in the normal hand ironing of fabrics.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Class of flatirons required (see 1.2).
(c) When representative sample is waived (see 3.1).
(d) Type of finish (see 3.5).
(e) Color of enamel finish (see 3.5.3).
(f) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

DOT waives coordination

Preparing activity:

GSA-FSS

Orders for this publication are to be placed with the General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 10 cents each.

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